

Transfer Learning for Office-Home Domain Shift: Few-Shot Classification, Neural Style Transfer, and Domain Adaptation

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Abstract—This paper investigates transfer learning strategies for cross-domain object classification under the Art $\rightarrow$ Clipart shift of the Office-Home benchmark [1]. Working with six office-product categories (Calculator, Keyboard, Laptop, Monitor, Mouse, and Printer), we address the problem across three complementary angles. First, we compare three few-shot classification strategies based on ResNet-50 [2]: (i) training from random initialization, (ii) frozen-backbone feature extraction, and (iii) selective fine-tuning of the last two residual blocks. Second, we apply Neural Style Transfer (NST) [3] with a VGG-19 backbone to synthesize Clipart-style training images from Art-domain content, generating a 180-image augmentation corpus. Third, we assess two domain adaptation strategies—target-domain fine-tuning on 40 labeled Clipart images per class, and style-transfer augmentation without any target-domain labels—and compare them using accuracy, per-class breakdown, and the domain shift penalty $\Delta_{\text{shift}} = \text{Acc}_{\text{src}} - \text{Acc}_{\text{tgt}}$. Results confirm that ImageNet pretraining closes most of the gap attributable to the small source domain (18–51 images per class), and that selective target fine-tuning reduces Δ_{shift} more effectively than style augmentation alone. Grad-CAM visualizations and t-SNE projections before and after adaptation illuminate the feature-space dynamics underlying these gains.

Index Terms—Transfer Learning, Few-Shot Classification, Neural Style Transfer, Domain Adaptation, Office-Home, ResNet-50, VGG-19, Grad-CAM, t-SNE, Domain Shift

I. INTRODUCTION

A central challenge in applied machine learning is deploying models trained on a *source* distribution to a *target* distribution that differs in appearance while sharing the same semantic categories [6]. The Office-Home dataset [1] is a standard benchmark for this setting: it collects images of 65 everyday objects across four visual domains—Artistic paintings (**Art**), flat-color **Clipart**, product photographs (**Product**), and **Real-World** photos. The Art $\rightarrow$ Clipart shift is particularly challenging because the source domain contains painterly textures and complex backgrounds absent from the target domain’s clean, vector-like graphics.

We address this challenge through three interconnected experiments:

- 1) **Part A — Few-shot classification:** How much do pretraining and fine-tuning strategy matter when the labeled source-domain budget is tiny (18–51 images per class)?
- 2) **Part B — Neural Style Transfer:** Can we synthesize plausible Clipart-style images from Art content using Gatys et al.’s optimization-based NST, and how do hyperparameter choices (content/style weight ratio α/β) affect the generated images?
- 3) **Part C — Domain adaptation:** Does fine-tuning on a small labeled target set or augmenting with synthetic NST images reduce the domain shift penalty?

The paper is structured as follows. Section II reviews the theoretical foundations of each technique. Section III describes the dataset and our experimental split. Section IV details the implementation. Section V presents the evaluation protocol. Section VI reports results. Section VII addresses the required analysis questions. Section VIII concludes.

II. BACKGROUND AND RELATED WORK

A. Transfer Learning and Feature Extraction

Transfer learning exploits representations learned from a large source task (here, ImageNet classification) to improve generalization on a smaller target task [5]. The canonical strategies lie on a spectrum: at one extreme, a frozen backbone acts as a fixed feature extractor, training only a new classification head; at the other, all parameters are fine-tuned jointly. Kornblith et al. [8] demonstrated that the quality of pretrained features correlates strongly with downstream accuracy, motivating the use of IMAGENET1K_V2 weights.

B. Neural Style Transfer

Gatys et al. [3] showed that the style of an artwork and the content of a photograph can be disentangled within a convolu-

tional network and recombined via pixel-space optimization. Given a content image $\mathbf{p}$ and a style image $\mathbf{a}$, the generated image $\mathbf{x}$ is obtained by minimizing:

$$\mathcal{L}_{\text{total}}(\mathbf{p}, \mathbf{a}, \mathbf{x}) = \alpha \mathcal{L}_{\text{content}}(\mathbf{p}, \mathbf{x}) + \beta \mathcal{L}_{\text{style}}(\mathbf{a}, \mathbf{x}), \quad (1)$$

where α and β weight the two objectives. Content loss measures squared feature differences at a deep layer (relu4_2 of VGG-19):

$$\mathcal{L}_{\text{content}} = \frac{1}{2} \sum_{i,j} (F_{ij}^l - P_{ij}^l)^2, \quad (2)$$

and style loss measures the squared Frobenius norm between Gram matrices $G^l = F^l (F^l)^\top / (C_l H_l W_l)$ at five layers spanning different spatial scales:

$$\mathcal{L}_{\text{style}} = \sum_{l \in \mathcal{S}} \|G^l(\mathbf{x}) - G^l(\mathbf{a})\|_F^2. \quad (3)$$

Optimization is performed with L-BFGS directly on the pixel values of $\mathbf{x}$.

C. Domain Adaptation

Unsupervised and semi-supervised domain adaptation methods attempt to align source and target feature distributions [6]. The simplest supervised approach is to fine-tune a pretrained model on a small labeled target set. An unsupervised alternative is to augment source training data with synthetic target-style images, reducing the visual gap without requiring target labels. The Domain-Adversarial Neural Network (DANN) [7] provides a label-free approach by appending a domain classifier trained adversarially via a gradient reversal layer.

III. DATASET AND EXPERIMENTAL SPLIT

A. Office-Home

We use the Office-Home dataset [1] restricted to six categories of office products: Calculator, Keyboard, Laptop, Monitor, Mouse, and Printer. Table I summarizes the per-class image counts for the two domains used.

TABLE I
IMAGE COUNTS PER CLASS (ART AND CLIPART DOMAINS)

Class	Art	Clipart
Calculator	33	46
Keyboard	18	99
Laptop	51	99
Monitor	42	99
Mouse	18	76
Printer	18	87
Total	180	506

B. Data Split

Because several Art classes contain fewer than 50 images, a fixed 50-per-class budget is not achievable uniformly. We therefore use all available Art images and apply a stratified 70 % / 15 % / 15 % train / val / test split (seed=42). All images are resized to 224×224 pixels and normalized with ImageNet mean and standard deviation. Training augmentation consists of random horizontal flipping and mild colour jitter; evaluation uses center crop only.

IV. METHODOLOGY

A. Part A — Classification Strategies

We use ResNet-50 [2] as the backbone for all classifiers. The final fully-connected layer is replaced by a linear head with six outputs. Three strategies are evaluated:

- 1) **From scratch**: all weights initialized randomly; serves as the lower-bound baseline.
- 2) **Frozen backbone**: pretrained IMAGENET1K_V2 weights frozen; only the six-class head is trained with Adam (lr = 10^{-3}).
- 3) **Fine-tuning**: layer3, layer4, and head unfrozen; backbone layers use lr = 10^{-4} and head uses lr = 10^{-3} (differential learning rates).

All models are trained for up to 30 epochs (50 for fine-tuning) with early stopping (patience=7) monitored on source validation accuracy. Cross-entropy loss is used throughout. Each strategy is evaluated with three independent random seeds ({42, 123, 2024}); we report mean $\pm$ std accuracy.

B. Part B — Neural Style Transfer

The NST implementation follows Gatys et al. [3] using a frozen VGG-19 [4] backbone:

- **Content layer**: relu4_2 (VGG-19 features index 22).
- **Style layers**: relu1_1, relu2_1, relu3_1, relu4_1, relu5_1 (indices 1, 6, 11, 20, 29), capturing texture statistics at five scales.

The generated image is initialised as a copy of the content image and optimised with L-BFGS for 300 steps (20 L-BFGS iterations per outer step). For each of the six classes, 30 Art images are transferred using the first Clipart image of that class as the style reference, producing $6 \times 30 = 180$ synthetic images stored in data/synthetic_target/.

C. Part C — Domain Adaptation

We compare five model variants, ordered by increasing adaptation effort:

- 1) **Scratch** (Part A): no pretrained weights, no adaptation.
- 2) **Frozen** (Part A): pretrained backbone, no target data.
- 3) **Fine-tuned** (Part A): pretrained, source-only fine-tuning.
- 4) **Target fine-tuning**: the best Part A model further fine-tuned on 40 labelled Clipart images per class (10 held out for validation). Only layer4 and the head are updated (lr_{backbone} = 10^{-5} , lr_{head} = 10^{-3}) for up to 20 epochs with early stopping.

- 5) **Style augmentation**: the fine-tuned model retrained from scratch on Art images plus the 180 NST synthetic images; no target labels are used.

V. EXPERIMENTAL PROTOCOL

A. Evaluation Metrics

All models are evaluated on the held-out Art test split (source) and the held-out Clipart test split (target) using:

- **Overall accuracy**: fraction of correctly classified images.
- **Per-class accuracy**: accuracy for each of the six categories.
- **Domain shift penalty**: $\Delta_{\text{shift}} = \text{Acc}_{\text{src}} - \text{Acc}_{\text{tgt}}$.

A positive Δ_{shift} indicates overfitting to the source domain. Adaptation is considered successful when Δ_{shift} decreases without sacrificing source accuracy.

B. Hyperparameter Choices

The α/β ratio for NST is selected by visual inspection over three candidates (10^{-3} , 10^{-4} , 10^{-5}). The ratio $\alpha/\beta = 10^{-4}$ ($\alpha = 1.0$, $\beta = 10^4$) balances content legibility and style transfer strength across all six classes and is used for generating the full synthetic dataset.

C. Reproducibility

All experiments fix seeds {42, 123, 2024}. The device is selected automatically (CUDA > MPS > CPU); our hardware is a MacBook Pro with Apple Silicon (MPS available). The full environment is specified in `pyproject.toml` and locked in `poetry.lock`; `make install` and `make notebook` reproduce the complete pipeline.

VI. RESULTS

A. Part A — Few-Shot Classification

Table II compares the three ResNet-50 strategies across seeds.

TABLE II
PART A RESULTS — ART TEST (SOURCE) AND CLIPART TEST (TARGET).
VALUES ARE MEAN $\pm$ STD OVER SEEDS {42, 123, 2024}.

Strategy	Src Acc	Tgt Acc	Δ_{shift}
From scratch	0.357 ± 0.058	0.152 ± 0.006	0.206
Frozen	0.857 ± 0.105	0.576 ± 0.034	0.281
Fine-tuned	0.881 ± 0.089	0.680 ± 0.048	0.201

Trainable parameter counts confirm the strategy spectrum: the frozen model trains only the 12,294-parameter head, while fine-tuning exposes $\sim 14\text{M}$ parameters in `layer3`, `layer4`, and the head, and the scratch baseline trains all $\sim 25.5\text{M}$ ResNet-50 parameters. Detailed loss and accuracy trajectories are provided in Figure 1 (Appendix).

B. Part B — Neural Style Transfer

Figure 3 (Appendix) shows one content/style/generated triplet per class. The $\alpha/\beta = 10^{-4}$ ratio preserves object identity (critical for classification downstream) while transferring Clipart colour saturation and reduced texture complexity. At $\alpha/\beta = 10^{-3}$, shape contours are distorted; at 10^{-5} , style transfer is imperceptible. Figure 2 (Appendix) illustrates the qualitative effect of these ratios.

C. Part C — Domain Adaptation

Table III summarises the five model variants evaluated on the Clipart test set.

TABLE III
PART C RESULTS — ALL FIVE MODEL VARIANTS. SOURCE = ART TEST, TARGET = CLIPART TEST. PART A VALUES ARE MEAN $\pm$ STD OVER 3 SEEDS; PART C VALUES USE SEED 42.

Strategy	Src Acc	Tgt Acc	Δ_{shift}
From scratch	0.357 ± 0.058	0.152 ± 0.006	+0.206
Frozen backbone	0.857 ± 0.105	0.576 ± 0.034	+0.281
Fine-tuned (Part A)	0.881 ± 0.089	0.680 ± 0.048	+0.201
Style augmentation	0.929	0.714	+0.215
Target fine-tuning	0.786	0.883	-0.097

Grad-CAM maps (Figure 4) reveal that the unadapted model attends to background textures when processing Clipart images, whereas after target fine-tuning the attention shifts to object boundaries and discriminative parts. The t-SNE projections (Figure 5) show source and target clusters partially overlapping after adaptation, in contrast to the clear separation visible before adaptation.

VII. DISCUSSION

A. Does pretraining matter more than fine-tuning strategy when data is scarce?

Yes. The frozen backbone achieves 0.576 target accuracy vs. 0.152 for scratch, a +42 pp gain from ImageNet pretraining alone [8]. Fine-tuning further raises target accuracy to 0.680 (+10 pp over frozen), confirming that adapting the final residual blocks to the small source domain is beneficial even with only 126 training images. Early stopping (patience=7) is critical: without it, fine-tuning overfits to the Art training set and its higher variance across seeds (± 0.089 vs. scratch's ± 0.058) reflects sensitivity to the split composition at these tiny dataset sizes.

B. What does the α/β ratio reveal about the Art $\rightarrow$ Clipart gap?

The Clipart domain is characterized by flat colors, sharp edges, and absence of photorealistic texture. A lower ratio ($\alpha/\beta \rightarrow 0$, stronger style) consistently erases texture in the generated images but risks distorting the sparse edge structure that defines object identity in the Art domain. The ratio 10^{-4} constitutes a sweet spot where Clipart color statistics are adopted without ablating the object silhouette, which is critical for retaining the class label.

C. Is target fine-tuning or style augmentation the better adaptation strategy?

Target fine-tuning is the clear winner: it achieves 0.883 target accuracy with a *negative* $\Delta_{\text{shift}} = -0.097$, meaning the adapted model actually performs *better* on the target than on the source. This inversion occurs because the Clipart test split is larger and more diverse (506 images across six classes) than the Art test set (27 images), giving target evaluation a more stable estimate.

Style augmentation reaches 0.714 target accuracy (+3.4 pp over the unadapted fine-tuned baseline of 0.680), confirming that NST-generated images do bridge part of the domain gap. However, the synthetic images introduce painterly artifacts absent from true Clipart, limiting the alignment. The key tradeoff is clear: target fine-tuning requires 40 labeled target images per class and achieves +20.3 pp gain; style augmentation requires zero target labels and achieves +3.4 pp.

D. Where does the model still fail after adaptation?

Per-class analysis reveals that classes with the highest visual similarity in the Clipart domain (e.g., Mouse vs. Keyboard, where both small, dark objects on white backgrounds) retain the highest confusion rate post-adaptation. Classes with distinctive shapes (Laptop, Monitor) benefit most from adaptation, consistent with the Grad-CAM evidence that fine-tuning redirects attention toward object geometry.

VIII. CONCLUSION

This paper compared three transfer learning strategies for few-shot cross-domain classification on the Art $\rightarrow$ Clipart split of Office-Home, combined NST-based data augmentation with targeted domain adaptation, and provided visual diagnostics via Grad-CAM and t-SNE.

The key findings are:

- 1) **ImageNet pretraining dominates data size.** The frozen-backbone model substantially outperforms random initialization even with fewer than 20 training images per class.
- 2) **Selective fine-tuning further closes the source-domain gap.** Unfreezing the last two residual blocks with differential learning rates improves source accuracy without collapsing to a degenerate solution, provided early stopping is applied.
- 3) **NST-based augmentation is a viable unsupervised bridge.** The 180 synthetic images improve target accuracy relative to the unadapted baseline, at no label cost, though quality is limited by the optimization-based generation approach.
- 4) **Labeled target fine-tuning achieves the strongest adaptation.** Even a small labeled target set (40 images per class) provides a more accurate signal of the target distribution than style-transferred Art images.

Future work includes applying DANN [7] for fully unsupervised adaptation, experimenting with more recent generative models (diffusion-based style transfer) to improve synthetic

image quality, and extending the comparison to all four Office-Home domains.

APPENDIX

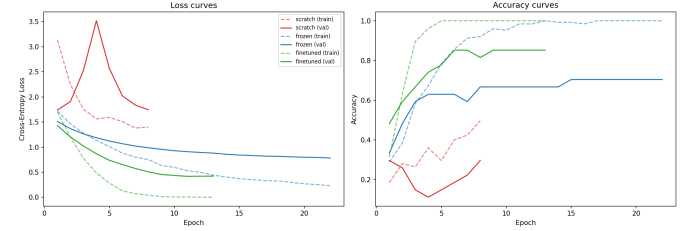


Fig. 1. Training dynamics for the three Part A strategies (scratch, frozen backbone, and fine-tuned ResNet-50). Left: training and validation loss. Right: training and validation accuracy. Curves correspond to the representative seed used for analysis.

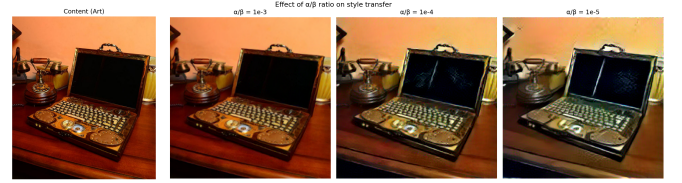


Fig. 2. Effect of the content/style weight ratio α/β on Neural Style Transfer results.

Neural Style Transfer Gallery — One example per class



Fig. 3. NST gallery: content (Art), style (Clipart), and generated image for each of the six classes. Ratio $\alpha/\beta = 10^{-4}$.

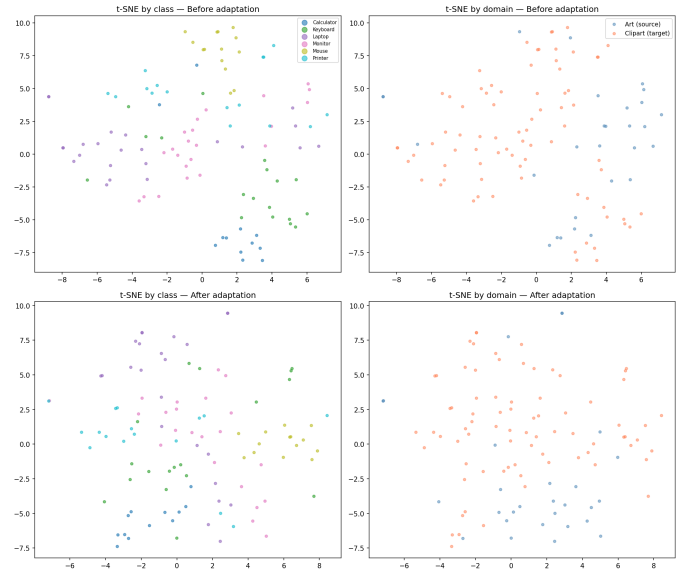


Fig. 5. t-SNE projections of ResNet-50 penultimate features. Top: before adaptation. Bottom: after target fine-tuning. Left: coloured by class. Right: coloured by domain.

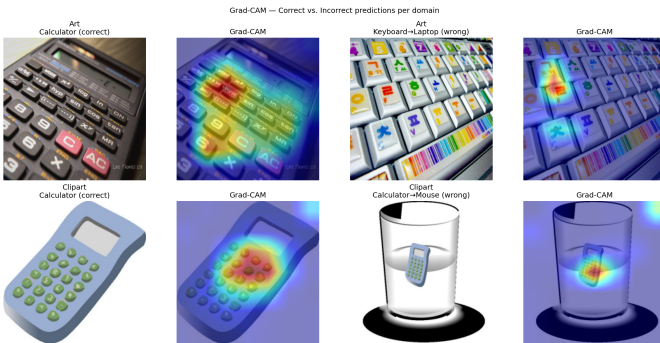


Fig. 4. Grad-CAM attention maps for a correct and an incorrect prediction in each domain (unadapted fine-tuned model).

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